

Encapsulation of hydrophilic compounds in small extracellular vesicles: Loading capacity and impact on vesicle functions

Advanced Healthcare Materials 2022

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Journal: Advanced Healthcare Materials

Year: 2022

Citation as given: Adv Healthc Mater. 2022

DOI: 10.1002/adhm.202100047 | **Link:** <https://doi.org/10.1002/adhm.202100047>

Abstract

Their natural functions in intercellular communication render extracellular vesicles (EV) highly attractive for drug delivery applications. However, the loading efficiency of present methods to incorporate particularly hydrophilic low molecular weight drugs of biomedical interest is largely unexplored, as is the impact these methods may have on the intrinsic structural and biological vesicle properties. Here, different methods are exploited to incorporate hydrophilic non-membrane permeable compounds into stem cell-derived small EV, and to assess the vesicle characteristics after the different loading processes. When comparing several methods head-to-head, the loading capacity increases in the order saponin $\leq$ sonication $<$ fusion $<$ freeze-thawing $\leq$ osmotic shock. Interestingly, the structural and biological functions of small EV are dependent on the applied encapsulation process, with the functional properties being altered at a greater extent. Therefore, the importance of including additional characterization parameters to probe alterations of the biological functionality of small EV is clearly demonstrated. Here, freeze-thawing and particularly the osmotic shock have proven to be the most appropriate methods for EV loading, as they achieve a high drug encapsulation and yet preserve the investigated structural and biological vesicle characteristics.